Address AI Bias with Fairness Criteria

Individual Fairness metrics

- **Distance-based Fairness:** similar individuals—measured by proximity in input space—should receive similar predictions or outcomes,
  - often enforced via Lipschitz constraints or distance metrics between feature vectors.
  - A function f is K-Lipschitz if $\| f(x)-f(y) \| \leq K \| x-y \|$ for all inputs x, y, where K is a constant (K=1 for strict bounds).
- **Counterfactual Fairness:** A model is counterfactually fair if its prediction for an individual remains unchanged when a protected attribute (e.g., race, gender) is hypothetically altered while keeping all other factors constant.

- **Similarity-based Fairness via Embeddings:** Uses learned representations (embeddings) to measure similarity between individuals
  - and enforces equal predictions for those close in the embedding space.
  - Embeddings extend distance-based by generalizing to non-Euclidean similarities.

  Ex: Two resumes with near-identical skills but slight word variations. BERT can be used for semantic similarity.

Group Fairness metrics

Idea: compare the outcomes of AI algorithms by comparing the statistics across groups.

Notations: A: group identity of the subject, Y: the actual class, $\hat{Y}$: the outcome of the model.

1. Treatment Equality: focuses on balancing the ratio of false positive rate to false negative rate across different groups. Ensures the burden of mistakes is shared equally.

Mathematically,

$$\Pr(\hat{Y}=0 \mid Y=1, A=0) / \Pr(\hat{Y}=1 \mid Y=0, A=0) = \Pr(\hat{Y}=0 \mid Y=1, A=1) / \Pr(\hat{Y}=1 \mid Y=0, A=1)$$

The ratio FN rate/FP rate is same across groups.

Group Fairness metrics

Suppose that there are 100 loan applicants from group *A* and 50 from group B.

For A, 8 were wrongly denied a loan (FN_A) and another 6 were wrongly approved (FP_A). The remaining predictions were true, so $TP_A + TN_A = 86$.

For group B, 5 were wrongly denied (FN_B) and 2 were wrongly approved (FP_B). The remaining predictions were true, so $TP_B + TN_B = 43$.

$$ACC_A = (86)/(86+ 8 + 6) = 0.86$$

$$ACC_B = (43)/(43 + 5 + 2) = 0.86$$

The ratio of false negatives to false positives equals 8/6 = 1.33 for A and 5/2 = 2.5 for B.

Hence, TE = 2.5 - 1.33 = 1.167, even though both groups have the same accuracy.

Group Fairness metrics

2. **Equalized Odds:** requires that a predictor is conditionally independent of the protected attribute A, given the true outcome Y. Mathematically, this can be expressed as:

$$\Pr(\hat{Y}=1|A=0, Y=y) = \Pr(\hat{Y}=1|A=1, Y=y) \quad \forall y \in \{0,1\}.$$

TP and FP rates must be equal across both protected and unprotected groups.

Try: Update the numbers in the confusion matrix from the previous example to make the classification satisfying equalized odds. What is the accuracy now?

Limitations: Equality of odds is difficult to achieve in practice, as it requires balancing multiple rates. It also may conflict with other fairness criteria and overall accuracy.

Group Fairness metrics

3. Statistical Parity: A predictor satisfies statistical parity (also called demographic parity) if it ensures that **individuals from different groups have the same probability of receiving a positive outcome**. Mathematically,

$$\Pr(\hat{Y}=1|A=0) = \Pr(\hat{Y}=1|A=1)$$

TP rate+FP rate is same across groups.

Limitations: This metric does not take into account the differences in true qualification rates across groups.

Use cases: Hiring algorithms to ensure equal hiring rates across genders and loan approval systems to provide equal approval rates across different ethnicities.

If base rates differ between sensitive groups, a classifier cannot satisfy equalised odds and demographic parity.

Group Fairness metrics

4. Equal Opportunity: A predictor satisfies equal opportunity fairness condition if

$$\Pr(\hat{Y}=1|A=0, Y=1) = \Pr(\hat{Y}=1|A=1, Y=1)$$

Ensures that **qualified individuals from different groups don't receive unfair discrimination.**

TP rate must be equal across groups.

Use Cases: Equally qualified applicants (e.g., those who will repay the loan) are equally likely to be approved the loan across groups.

Limitations: It requires accurate measurement of qualification, which can be subjective, and it may not address biases in the qualification assessment itself.

Group Fairness metrics

5. Accuracy Equality: A predictor satisfies accuracy equality if the classification accuracy is the same for all subgroups.

$$\Pr(\hat{Y}=y|A=0) = \Pr(\hat{Y}=y|A=1) \quad \forall y \in \{0,1\}.$$

TP and TN rates are equal.

This means the predictor makes correct predictions at the same rate for all groups, regardless of whether the prediction is positive or negative.

5. Predictive Equality: A predictor satisfies predictive equality (predictive parity) if the predicted positive outcome has the **same precision** across different groups.

$$\Pr(Y=1|A=0, \hat{Y}=1) = \Pr(Y=1|A=1, \hat{Y}=1)$$

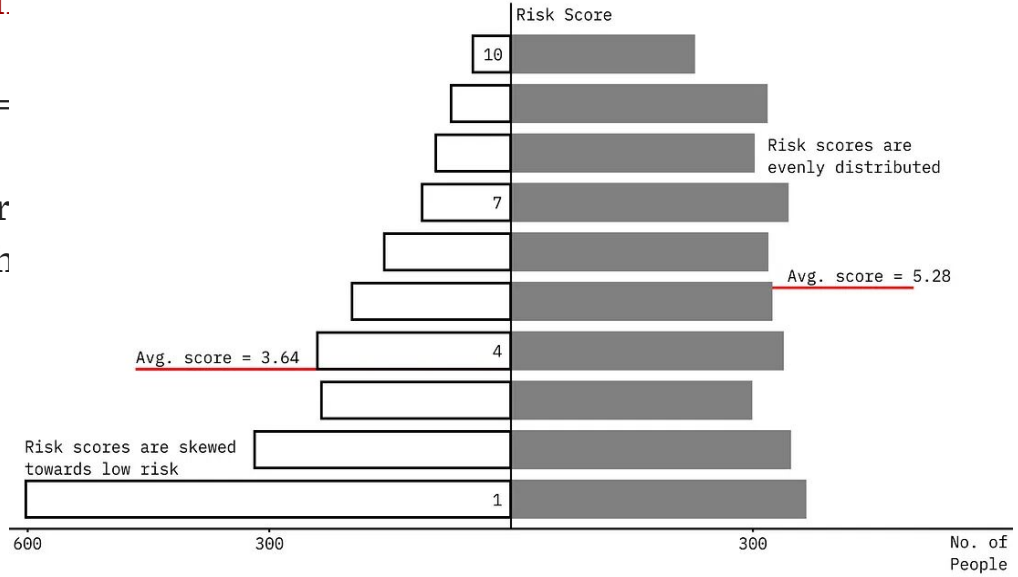
Group Fairness metrics

7. Calibration across groups (for score based predictors): A probabilistic/score based predictor $Y \in [0,1]$ satisfies calibration across groups if for every predicted score p ,

Is the predicted probability equally rel

$$\Pr(\hat{Y}=1|A=a, S=s) = \Pr(\hat{Y}=1|A=$$

Individuals with the same predicted pr
being classified in the positive class wh
unprotected group.



https://medium.com/@lamdaa/compas-unfair-algorithm-812702ed6a6a

Measuring bias is challenging

Theorem (Kleinberg et al., 2016, Chouldechova 2017**):** If base rates differ between groups (i.e., $Pr(Y=1|A=a) \neq Pr(Y=1|A=b)$, and the predictor is not perfect (i.e., there is predictive uncertainty), then it is **impossible** for any scoring function to simultaneously satisfy

1) calibration across groups,
2) balance for the positive class (among those **with positive labels**, the **average** predicted risk is equal across groups) and,
3) balance for the negative class (among those **with negative labels**, the **average** predicted risk is equal across groups).

Weerts, H. J. P. (2025). Decoding Algorithmic Fairness: Towards Interdisciplinary Understanding of Fairness and Discrimination in Algorithmic Decision-Making. [Phd Thesis 1 (Research TU/e / Graduation TU/e), Mathematics and Computer Science]. Eindhoven University of Technology.

Kleinberg, J., Mullainathan, S., & Raghavan, M. (2016). Inherent trade-offs in the fair determination of risk scores. arXiv preprint arXiv:1609.05807. https://arxiv.org/abs/1609.05807

Mitigation Techniques

Overview of Bias Mitigation

1. Data-Level Interventions (Sampling/re-weighting)
2. Preprocessing: Modify dataset before training (Feature transformation/removal)
3. In-processing: Incorporate fairness during training (modified optimization function)
4. Post-processing: Adjust outputs after training (Threshold Adjustment, re-ranking)
5. Beyond metrics

Mitigation Techniques

Data-Level Interventions

1. Balanced Sampling / Reweighting

Random oversampling (duplicating minority examples) or undersampling (removing majority examples) to achieve equal class proportions per training batch.

If the group population is sparse then **each sample from that group is given higher weight** resulting in debiasing the data.

Adding weights to sample i from group a_i as $w_i = 1/Pr(A = a_i)$.

The loss function is then modified to $\sum wi \cdot \ell(yi, \hat{y}i)$.

2. Datasheets for Datasets

Proposed by Timnit Gebru (co-authored the ‘Gender Shades’ paper) and colleagues in 2018.
A datasheet typically covers:

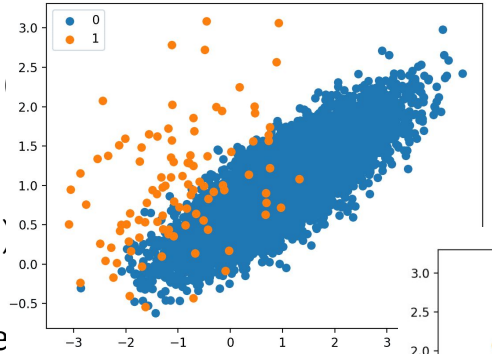
- Motivation: Why created, by whom, for what task.
- Composition: Size, features, demographics, errors or noise.
- Collection Process: Methods, time/place, incentives, consent.
- Recommended Uses and Limitations: Intended tasks, known biases, maintenance plans.

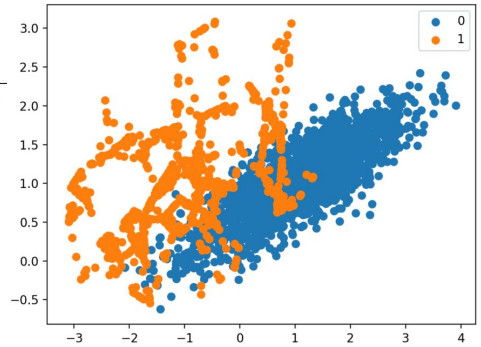
3. Synthetic Data Generation

These techniques ensure equal representation from minority groups by appropriately modifying the data sampling by

Synthetic Minority Over-sampling Technique (

1. Pick a sample from the minority class.
2. Find its k nearest neighbors (default k=5)
3. Choose one randomly.
4. Create a new synthetic point somewhere





Question 1: In a fraud detection model using SMOTE-balanced data, demographic parity ensures equal positive rates across genders, but it might lower overall accuracy. Is enforcing parity always worth the trade-off for a bank, or should accuracy come first?

Question 2: Datasheets reveal your image dataset underrepresents South Asian faces due to web-scraping bias. Should you oversample with SMOTE to fix outcomes (group parity) or avoid sensitive proxies like location in preprocessing? What risks does each hide?